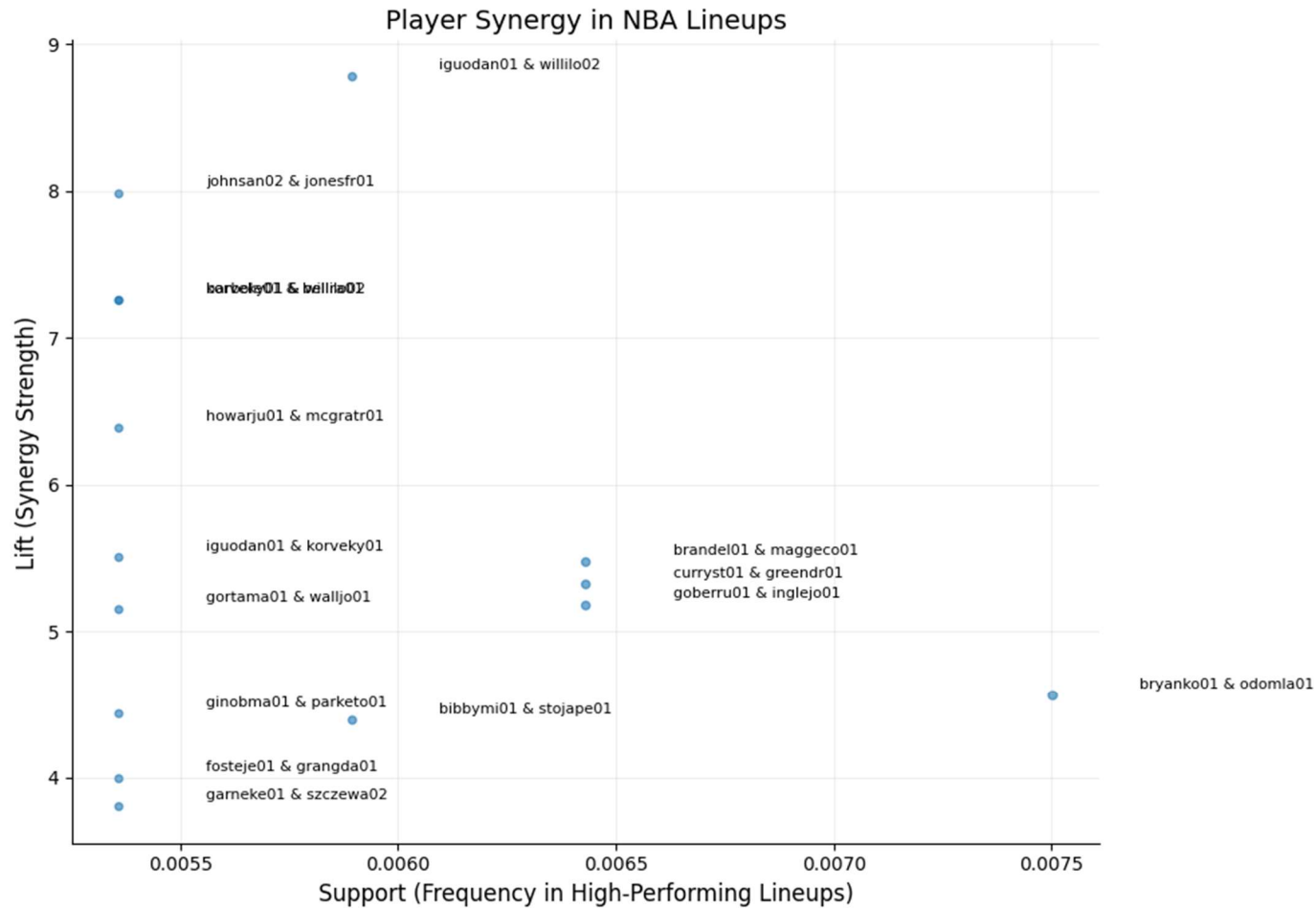


Mining Player Synergy in NBA Lineups



Legend

Each point represents a pair of players. The x-axis shows support, defined as the frequency of the pair in high-performing lineups. The y-axis shows lift, which measures how much more frequently the pair appears compared to expectation under independence. The size of each point is proportional to its support.

Findings

- ① Several player pairs exhibit strong positive synergy, with lift values significantly greater than 1, indicating non-random co-occurrence in high-performing lineups.
- ② The most effective pairs are located in the upper-right region of the plot, where both support and lift are relatively high, suggesting that these combinations are both frequent and impactful.
- ③ Many high-synergy pairs correspond to complementary roles, such as playmaker–finisher or shooter–screen setter combinations (e.g., Curry–Green, Gobert–Ingles).

④ Some pairs show very high lift but relatively low support, indicating strong but less frequently observed combinations.

⑤ Overall, the results suggest that team performance is influenced more by player interactions than by individual contributions alone.

Data and Methods

We used a publicly available NBA play-by-play dataset (1999–2022) with imputed on-court lineups. Each record contains detailed game events along with the five players on the court for both teams at every play. This enables direct analysis of lineup-level performance using only publicly accessible data.

To construct the analysis dataset, plays were grouped by unique player combinations. Each lineup was treated as a unit of analysis, and its performance was evaluated using points per play (PPP), defined as the total points scored divided by the number of events. Although PPP is an approximation of offensive efficiency, it provides a consistent and computationally simple measure across all lineups.

Lineups with very low frequency were filtered out to reduce noise and improve statistical reliability. The remaining lineups were then divided into high-, mid-, and low-performing groups based on PPP quantiles, allowing us to focus on the most effective combinations.

To identify synergistic player relationships, we applied association rule mining. Each lineup was modeled as a transaction, and individual players were treated as items. We focused on pairwise combinations (size-2 itemsets) to mitigate sparsity and avoid the combinatorial explosion associated with larger player groups.

For each player pair, support was calculated as the proportion of high-performing lineups in which the pair appears. To measure the strength of association, we used lift, which compares the observed co-occurrence of two players in high-performing lineups to the expected frequency under independence. This formulation ensures that lift captures true interaction effects rather than simple frequency.

Finally, low-support pairs were filtered out to avoid unstable estimates, and extreme values were excluded to reduce the impact of rare combinations. The resulting set of player pairs represents statistically meaningful and interpretable synergy patterns.

Significance

This study highlights the importance of interactions between players in determining team performance. While basketball analysis is often centered on individual statistics, our results demonstrate that effective lineups emerge from combinations of players whose skills complement one another.

The findings show that high-performing lineups are not necessarily composed of the highest-scoring or most individually dominant players. Instead, successful combinations often reflect functional roles, such as playmaker–finisher or shooter–screen setter partnerships. This suggests that team performance is inherently a collaborative phenomenon shaped by player interactions.

From a practical perspective, these insights can inform lineup construction and rotation strategies. Coaches and analysts may use similar approaches to identify underutilized player combinations or to better understand why certain lineups perform well in specific contexts.

More broadly, this project demonstrates how data mining techniques can uncover hidden structures in complex systems. The methodology used here is not limited to sports analytics; it can be applied to other domains where performance depends on interactions between multiple entities, such as team collaboration, recommendation systems, or social networks.

By combining publicly available data with interpretable statistical measures, this work shows that meaningful insights into collective behavior can be derived without access to proprietary tracking systems.

GitHub

<https://github.com/Seamus113/nba-synergy-data-mining>